DeepSeek-V2: A Strong, Economical, and Efficient MoE Language Model

236B total parameters · 21B activated per token · 128K context length

Multi-Head Latent Attention (MLA) + DeepSeekMoE

arXiv:2405.04434

The LLM Scaling Dilemma: Performance vs. Training Cost and Inference Speed

- Larger dense models improve capability but scale training costs super-linearly.
- Autoregressive generation is bottlenecked by the KV cache: full key–value vectors per head per token consume massive memory and I/O bandwidth.
- Compression tricks (GQA, MQA) shrink the cache but measurably hurt model quality.
- MoE promises sparse activation, yet communication overhead and load imbalance often erase the savings.
- Need: an architecture that is simultaneously cheaper to train, faster to generate from, and stronger at inference.

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in a Transformer Block

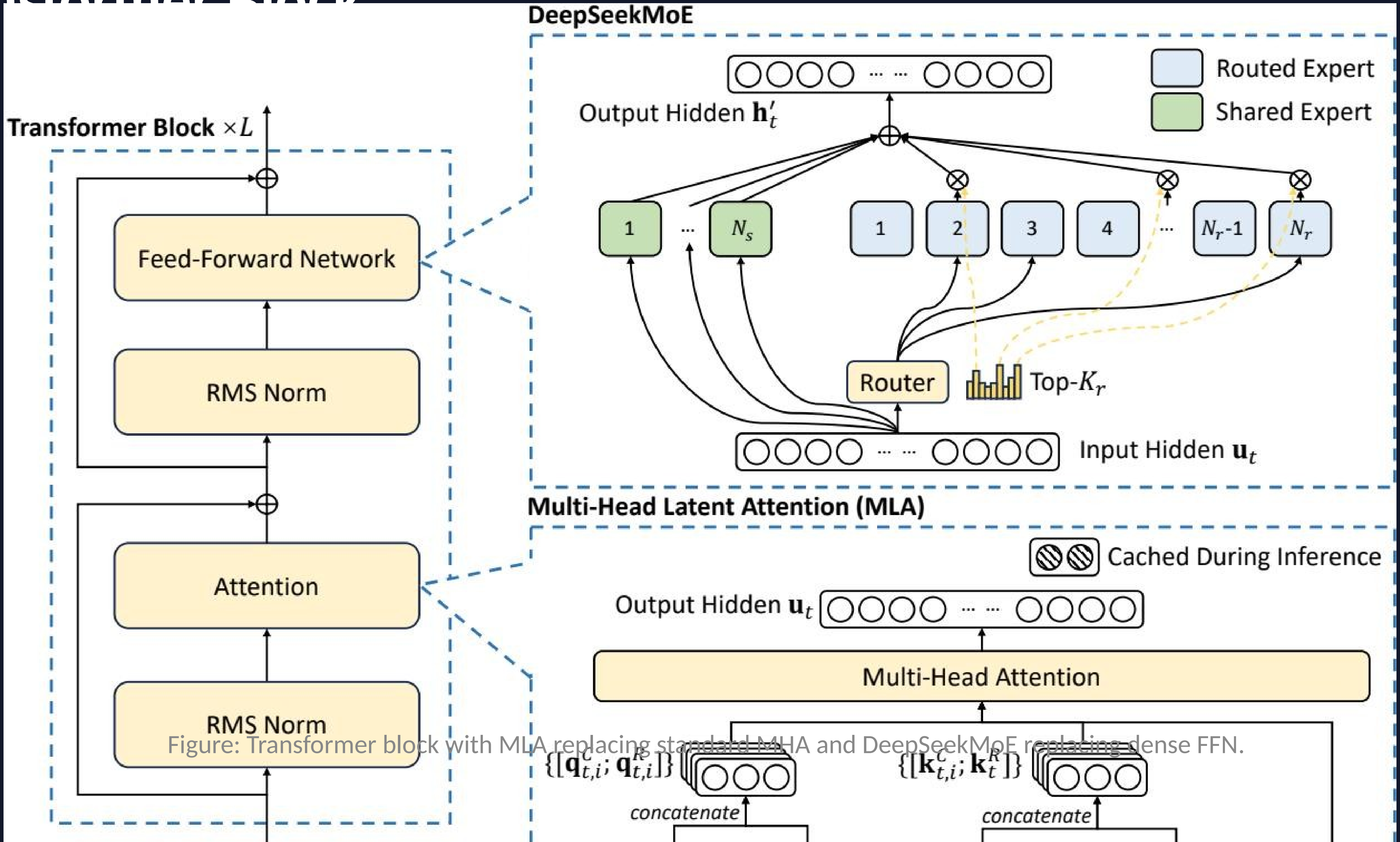


Figure: Transformer block with MLA replacing standard MHA and DeepSeekMoE replacing dense FFN.

Multi-Head Latent Attention Compresses KV Cache by 93.3%

- Low-rank key–value joint compression: cache a compact latent vector c_t^{KV} instead of full K/V matrices.
- Full keys and values are reconstructed on-the-fly via learned projection matrices.
- Decoupled RoPE-based positional key preserves positional information without inflating the cache.
- Result: better performance than standard MHA with dramatically less memory — strictly dominates GQA/MQA.

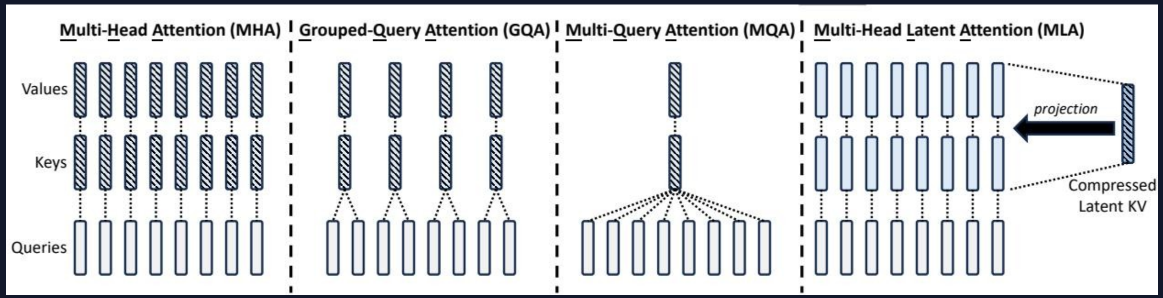


Figure: Attention mechanism comparison — MHA vs. GQA vs. MQA vs. MLA.

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

- Shared experts (N_s): always activated, capture common knowledge across all tokens.
- Routed experts (N_r): sparsely activated via a top-K_r router, enabling massive parameter counts with low per-token cost.
- Fine-grained expert segmentation increases specialization potential without adding parameters.
- Device-limited routing constrains each token to experts on at most M devices, cutting cross-node communication.
- Auxiliary losses enforce load balance across experts and devices; token dropping during training prevents overflow without affecting inference.

42.5% Training Cost Reduction and 5.76× Generation Throughput vs. DeepSeek 67B

Training Cost

42.5%
reduction

vs. DeepSeek 67B dense model

KV Cache Size

93.3%
reduction

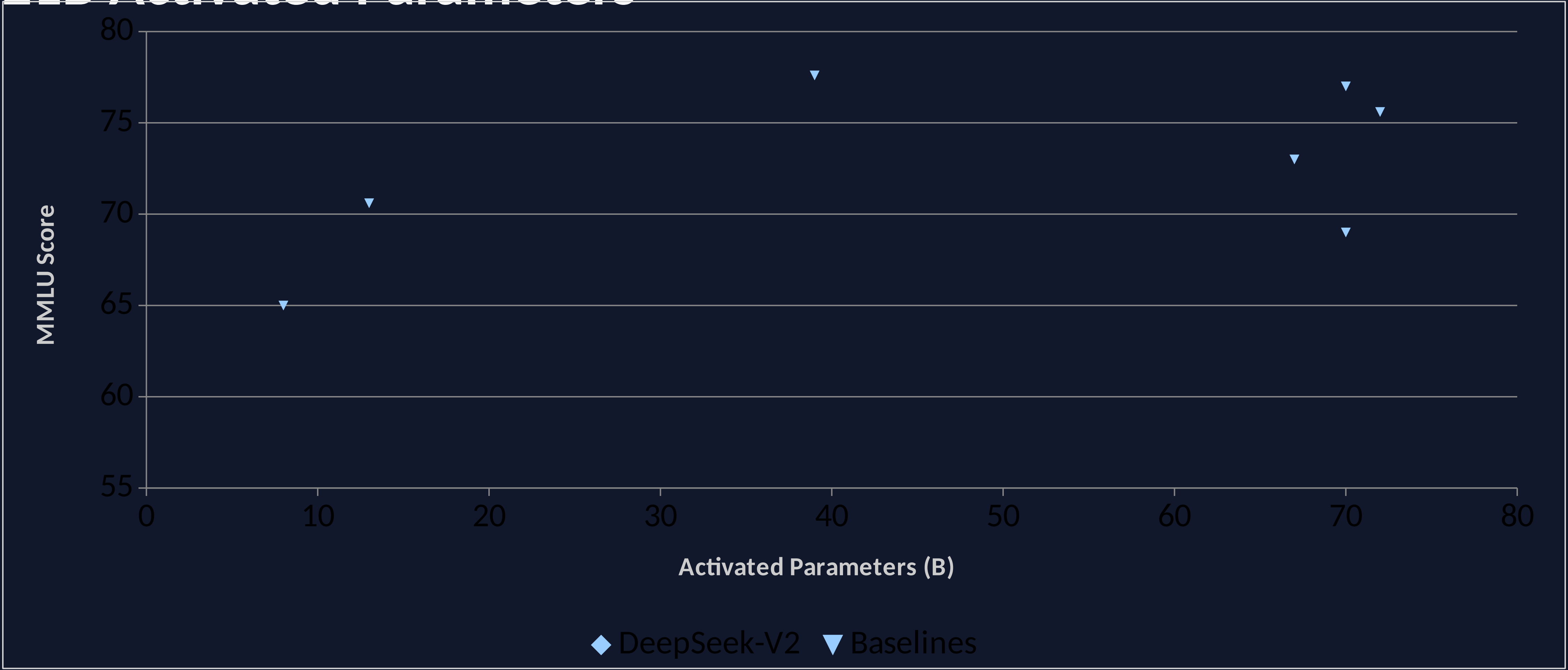
enabled by MLA latent compression

Generation Throughput

5.76×
higher

maximum throughput improvement

DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B Activated Parameters



DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

- Alignment pipeline: Supervised Fine-Tuning on 1.5M conversational sessions, followed by GRPO-based Reinforcement Learning.
- AlpacaEval 2.0 (length-controlled win rate): 38.9%
- MT-Bench (overall): 8.97
- AlignBench (overall, Chinese): 7.91
- Outperforms all open-source models and most closed-source models on Chinese open-ended evaluation.

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

1. MLA is a strictly better KV-cache solution: superior to MHA while cutting cache by 93.3%, dominating GQA/MQA.

2. DeepSeekMoE with fine-grained experts and shared-expert isolation trains a 236B model at 42.5% lower cost than a 67B dense model.

3. Only 21B activated parameters match or exceed 70B-class dense models on MMLU — Pareto-optimal efficiency.

4. 5.76× generation throughput improvement makes large-scale deployment practical.

5. DeepSeek-V2 Chat (RL) is the strongest open-source chat model in Chinese and competitive with closed-source systems in English.